

- 7 (a) There are quantitative and qualitative aspects of gravitational field that are analogous to those of electric field. Complete the table in Fig. 7.1 below to show these analogous aspects.

Analogy	
Gravitational Field	Electric Field
Mass	Electric Charge
Newton's Law of Gravitation	Coulomb's Law
Gravitational Field Strength is in the direction of decreasing Gravitational Potential	Electric Field Strength is in the direction of decreasing Electric Potential

Fig 7.1 [3]

- (b) In an imagined universe, Earth has a larger sibling Areth with which it forms a double planet system, and the two orbit about each other. Earth has a mass of 5.97×10^{24} kg and a radius of 6370 km. Areth has the same mean density as Earth but is 20% bigger in radius. They are separated by a distance of 96600 km.

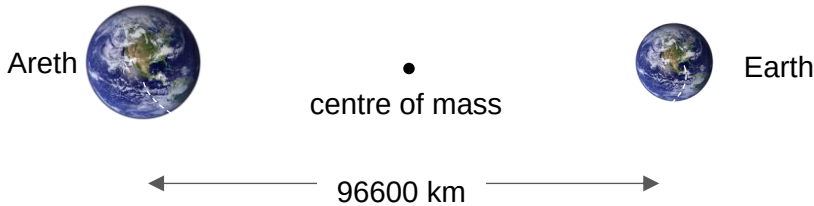


Fig. 7.2

- (i) Show that the magnitude of the force that Earth and Areth exert on each other is 4.40×10^{23} N.

Mass of Areth $M_A = 5.97 \times 10^{24} \times 1.20^3 = 1.03 \times 10^{25}$ kg [B1]

$$F = \frac{GM_E M_A}{d^2} = \frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 1.03 \times 10^{25}}{(9.66 \times 10^7)^2} \quad [B1]$$

$$= 4.40 \times 10^{23} \text{ N} \quad [2]$$

- (ii) Using Newton's Laws of Motion, explain why the two planets orbit about the centre of mass of the double planet system.

Each planet orbits due the gravitational force exerted by the other.

 As a system, there are no external forces exerted on them. [B1]

 Thus by Newton's 1st Law, the centre of mass must remain stationary. [B1]

 So the centre of mass is the centre of both planets' orbit. [2]

(iii) By considering the centripetal forces on the two planets, show that

$$\frac{\text{radius of Earth's orbit}}{\text{radius of Areth's orbit}} = 1.73$$

$$M_E r_E \omega_E^2 = M_A r_A \omega_A^2$$

$$\text{Since } \omega_E = \omega_A, \frac{r_E}{r_A} = \frac{M_A}{M_E} = \frac{1.03 \times 10^{25}}{5.97 \times 10^{24}} \text{ [B1]} = 1.73$$

[1]

(iv) Determine the orbital period of the double planet system in days.

$$r_E = \frac{1.73}{2.73} \times 9.66 \times 10^7 \text{ [B1]} = 6.12 \times 10^7 \text{ m}$$

$$F = M_E r_E \omega^2 \rightarrow 4.40 \times 10^{23} = 5.97 \times 10^{24} \times 6.12 \times 10^7 \times \omega^2 \text{ [M1]}$$

$$\omega = 3.47 \times 10^{-5} \text{ rad s}^{-1}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{3.47 \times 10^{-5}} \text{ [M1]}$$

$$= 1.81 \times 10^5 \text{ s} = 2.09 \text{ days [A1]}$$

period = days [4]

(c) P is a point on the surface of Earth that lies on the line joining the centres of Earth and Areth, as shown in Fig. 7.3.



Fig. 7.3

(i) State the value of the gravitational field strength at P due to Earth only.

field strength = 9.81 N kg⁻¹ [1]

- (ii) Using the values given in (b), calculate the gravitational field strength at P due to Areth only.

$$\text{Distance from Areth to P} = 9.66 \times 10^7 - 6.37 \times 10^6 = 9.02 \times 10^7 \text{ m}$$

[B1]

$$g = \frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 1.03 \times 10^{25}}{(9.02 \times 10^7)^2} \quad [\text{M1}]$$

$$= 0.0845 \text{ N kg}^{-1} \quad [\text{A1}]$$

field strength = N kg⁻¹ [3]

- (iii) Hence determine the gravitational force on a man of mass 80.0 kg standing at point P.

$$\text{Resultant field strength} = 9.81 - 0.0845 = 9.73 \text{ N kg}^{-1} \quad [\text{B1}]$$

$$\text{Gravitational force} = 80.0 \times 9.73 = 778 \text{ N} \quad [\text{A1}]$$

force = N [2]

- (iv) When the man in (iii) steps on a weighing scale, the reading produced is greater than the value determined in (iii). Suggest a reason for this.

The man is also orbiting about the double planet's centre of mass, thus it has an acceleration (and resultant force) towards it. [B1]

Thus the force the weighing scale exerts on the man (towards the centre of orbit) must be greater than the gravitational force (towards centre of Earth) [2]

[Total: 20]

8 (a) Apart from being different types of forces state a difference between electric force and